

Harsh Mohan Gajra

Senior Software Engineer

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Senior Software Engineer with 8+ years of solid experiences building for software development teams of industry cloud leaders like Amazon Web services (AWS) and dynamic AI data platform startups like Crux Data(backed by Goldman Sachs).

Highly experienced in all aspects of software development life-cycle and system design, from concept through development to delivery. Consistently recognized as a competent software engineer, with hard technical as well soft skills, required for coordinating cross-functional teams in a fast-paced environment to steer timely and quality project completion.

Equipped with a Master's in Computer Science from State University of New York at Binghamton, I am looking forward to obtaining the position of Technology Lead / Senior Software Engineer/ Staff engineer at a leading organization, to utilize my skills in software development to develop highly resilient and quality software products.

Education

Masters in Computer Science - State University Of New York, Binghamton, August 2016 - May 2018

Bachelor of Computer Engineering - K.J.Somaiya College of Engineering, Mumbai ,First class (2011- 2015)

Relevant Coursework: Data Structures and Algorithms, Database Systems, Operating Systems, Object Oriented Programming Languages, Design Patterns.

Technical Skills

Java 8 / 11	Amazon Web Services	GitHub
Spring Boot	Brazil Build system	Cloud Development kit
Git	JIRA	RDS
Postgres	MySQL	DynamoDB
AWS Systems Manager	SWF	AWS Lambda
CI/CD	EC2	AWS Cloudwatch
Airflow	Google Cloud Platform	BigQuery
Docker	Python 3	Pub/Sub
GPT	BigQuery	Git Actions
FastAPI	Spring Boot	Kubernetes

Work Experience

Senior Software Engineer – Crux Data (Feb 2023 – Present)

Technology Stack: Python 3, FastAPI, SQLAlchemy, Java, Spring Boot, OpenAI (LLMs), GCP, Docker, Kubernetes, Airflow, PostgreSQL, Cassandra, BigQuery, GitHub Actions, Apache Kafka

- Designed and developed event-driven, scalable APIs on Crux's Data Platform with FastAPI, Pub/Sub, and PostgreSQL—containerized in Docker, orchestrated via Kubernetes, and deployed on GCP for high availability and modularity.
- Architected a Git-driven CI/CD synchronization framework that converts declarative YAML workflows into backend domain models, automated via GitHub Actions to guarantee consistency between source and infrastructure.
- Developed an AI-powered adapter recommendation engine using OpenAI GPT to auto-configure ingestion pipelines in Apache Airflow, boosting onboarding speed and reducing manual tuning.
- Prototyped LLM-based pipelines to normalize heterogeneous file formats, improving data standardization and integration efficiency across downstream systems.
- Led the migration of 2 billion+ records from Cassandra to AlloyDB (PostgreSQL) using Python, Bash, SQL, Kubernetes, BigQuery, and GCP—realizing over \$300K in annual infrastructure savings.
- Built asynchronous Java APIs in Spring Boot with PostgreSQL and Kafka, deployed in Kubernetes on GCP, to support reliable messaging and microservice decoupling.

- Orchestrated and maintained complex data workflows on both GCP Cloud Composer and self-managed Airflow clusters in Kubernetes.
- Provided technical leadership and mentorship to a team of three junior and mid-level software engineers, fostering best practices and professional growth.

Software Engineer - Amazon Web Services, Data Migration Service (<https://aws.amazon.com/dms>)
 Seattle, Washington (Sept, 2021 – Feb 2023)

Technology Stack : Java, AWS services like DMS, RDS, CDK, S3, CloudFormation, Spring Boot

- Worked on designing and developing a Distributed Test Execution platform, that can manage and execute hundreds of thousands of test cases in parallel using serverless technologies like AWS Lambda, AWS Simple Workflow, AWS CloudWatch, Cloud development Kit and Java 11 to test different components AWS Data Migration Service like Control Plane, Data Plane(Kamino) and Serverless at scale.
- Worked on re-architecting the Canary and Test infrastructure to eliminate deployment bottlenecks and congestions leading to overall stabilization and faster deployments saving 50 hours a week.
- Started as 1st hire on the Platform STP team, mentored and lead a team of 3 interns and 2 engineers, and served as subject matter expert for multiple components of the software product.
- Wrote CI/CD pipelines using AWS Cloud development kit and CloudFormation to perform infrastructure deployments predictably and repeatedly, with rollback on error.
- Wrote automation, identified and eliminated bugs, fixed design flaws across the product stack concerning AWS DMS bringing down on call support tickets by 60 % and saving 2400 hours yearly.

Software Developer 2 – FINRA
 Reston, Virginia (May 2020 - Sept, 2021)

Technology stack: Java, AWS Services

- Contributed to the design and development of FINRA's Consolidated Audit Trail project.
- Wrote SQL queries to pull data and develop reports on stocks and options that were used in Project Dashboards.
- Collaborated with external teams and stakeholders to determine requirements and drive product deliverables.
- Designed and implemented features for FINRA's Consolidated Audit Trail project which manages an extensive trail of transactions across different stock exchanges.

Software Engineer - Pype Inc (Acquired by AutoDesk in May 2020)
 Herndon, VA, USA (July 2018 – May 2020)

Technology stack : Java 8, Spring Boot, MySQL, RabbitMQ

- Designed and developed internal software solutions for optimizing the business process followed by in-house Sales and Support teams thus saving the organization \$360,000 per year in costs and 15,000 man-hours per year.
- Designed the data model and developed the backend code for an Email Cadence software tool using Java 8, Spring Boot, JPA and MySQL database.
- Designed applications using the message brokers like RabbitMQ for message queuing and asynchronous job processing.
- Algorithm Development: Design, Analyze and implement algorithms in core Java for solving various use cases across the software products of the company.
- Wrote automations for the manual tasks performed by Customer Support and Success Teams saving our 1000 hours/year in the process.

Paper Publications

- **Paper:** Credit card Fraud Detection and Prevention Using Perceptron Training Algorithm.
- **Journal:** International Journal on Recent and Innovation Trends in Computing and Communication, Apr 2015
Link: <https://ijritcc.org/index.php/ijritcc/article/view/4182/4182>